



Thermophysical Assessment of High-Albedo Surfaces for Solar Radiation Management and Equivalent Energy Impact Evaluation

Alessia Di Giuseppe^{1*}, Federico Rossi^{1,2}

¹ CIRIAF (Interuniversity Research Centre on Pollution and Environment “Mauro Felli”), University of Perugia, 06125 Perugia, Italy

² Department of Engineering, University of Perugia, 06125 Perugia, Italy

* Correspondence: Alessia Di Giuseppe (alessia.digiuseppe@unipg.it)

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Abstract: The modification of surface radiative properties has been increasingly examined from the perspective of engineering thermophysics, particularly in relation to solar radiation management and its impact on surface energy balance. In this study, a methodological framework, referred to as the radiative forcing (RF)-meter, is introduced to quantify the RF variation induced by high-albedo surfaces under real environmental conditions. The proposed approach integrates ground-based irradiance measurements, astronomical calculations, and satellite observations to reconstruct the time-dependent radiative flux associated with albedo changes. Based on this formulation, the reduction in absorbed solar energy is evaluated and further interpreted in terms of equivalent energy impact, providing a consistent basis for linking radiative effects with thermophysical processes. The framework also allows the cumulative effect of RF reduction to be expressed in CO₂-equivalent terms, enabling a quantitative interpretation of long-term environmental and energy-related implications. Potential applications of the method are discussed with reference to high-irradiance regions, where surface reflectivity modification may contribute to both thermal regulation and energy demand reduction. The proposed methodology offers a structured approach for analysing surface energy behaviour within the context of engineering thermophysics, with possible extensions toward building energy efficiency and solar radiation control strategies.

Keywords: Engineering thermophysics; Radiative heat transfer; Solar radiation management; Surface albedo; Energy balance; Thermal performance; Radiative forcing

1 Introduction

This study proposes an operational methodology for assessing the climate mitigation potential of surface albedo enhancement.

Recent International Panel on Climate Change (IPCC) assessments emphasize that climate change mitigation requires the rapid deployment of multiple complementary strategies, supported by coherent regulatory and policy frameworks [1]. Among the main pathways commonly discussed in the literature are the expansion of renewable energy systems, improvements in energy efficiency, and the use of carbon capture and storage technologies. In parallel, market-based instruments such as Emissions Trading Schemes have been introduced in several jurisdictions to assign an economic value to greenhouse gas mitigation through cap-and-trade mechanisms [2]. Within these frameworks, the accounting unit is generally expressed in tonnes of CO₂-equivalent, allowing verified mitigation actions to be compared within a common metric.

Alongside conventional mitigation measures, the radiative properties of the Earth’s surface have also attracted attention. An increase in surface albedo reduces the fraction of incoming solar radiation absorbed by the surface-atmosphere system and can therefore generate a cooling effect. This physical link between albedo management and climate mitigation has been examined in several previous studies. Among the most widely investigated approaches are reflective materials and coatings for roofs, pavements, and other exposed surfaces in buildings and civil infrastructure, often referred to as cool materials [3]. In the present study, these interventions are grouped under the term High-Albedo Solutions (HAS), namely surface-based measures aimed at increasing terrestrial reflectivity and,

consequently, reducing positive radiative forcing (RF). In urban environments, HAS may also provide indirect benefits, including lower cooling-energy demand in buildings and mitigation of the Urban Heat Island effect, particularly in warm and cooling-dominated climates [4, 5].

Recent literature also suggests that the performance of reflective surfaces should not be assessed only in terms of peak summer cooling. Their effectiveness depends on climatic context, urban form, material properties, and ageing processes, and may vary substantially between roofs, façades, pavements, and mixed urban applications. In addition, while high albedo materials can reduce surface and near surface temperatures, their net benefit should also be evaluated with respect to possible trade-offs, such as seasonal heating penalties, changes in reflected shortwave radiation, and differences between building scale and neighborhood scale effects. These aspects indicate that albedo-based mitigation requires context sensitive assessment methods rather than static or purely nominal reflectance values [6–10].

Despite this recognized potential, a persistent methodological issue concerns the quantification of the climate benefit generated by albedo enhancement over time. Simplified estimates often neglect the temporal variability of solar irradiance, atmospheric transmissivity, material ageing, and the differences between continuous ground behaviour and discontinuous satellite observations. These aspects are crucial if HAS are to be evaluated not only as passive cooling measures, but also as interventions potentially eligible for climate accounting or crediting frameworks.

On this basis, the present work introduces the RF-meter, a methodology designed to reconstruct the time evolution of RF associated with an albedo increase by combining ground measurements, astronomical calculations, and satellite-based calibration. The final objective is to provide a more robust estimate of the CO₂-equivalent offset associated with HAS and to outline a possible pathway for their future inclusion in emission credit (EC) assessment procedures.

2 Methodological Framework of the Radiative Forcing-Meter

The climate benefit associated with HAS has been examined in the literature through a range of modelling approaches, most of which rely on the IPCC RF framework to relate surface albedo changes to an equivalent climatic effect.

A major challenge, however, lies in the accurate quantification of RF under real operating conditions. In practice, atmospheric transmissivity varies continuously along both the incoming and outgoing radiative paths because of changes in air mass, aerosol loading, and suspended particles. At the same time, surface albedo is not constant, since it may change over time because of weathering, dust accumulation, fouling, moisture, and material ageing. In addition, solar irradiance itself is inherently time dependent, being controlled by seasonality and by the diurnal evolution of solar position.

For this reason, the conversion of RF into CO₂-equivalent compensation requires a time resolved evaluation of the energy effect, based on the full temporal evolution of RF, consistently with the logic underlying the IPCC Global Warming Potential approach. Within this perspective, the present study introduces a dedicated methodology to estimate the CO₂ offset associated with HAS with greater temporal and physical resolution.

The proposed framework is intended to support the technical assessment of HAS in view of their possible recognition within EC procedures. The method integrates continuous ground-based measurements, analytical calculations, and satellite observations. More specifically, the RF-meter is designed to reconstruct the reduction in RF produced by a reflective intervention over time. This time dependent forcing signal can then be translated into an equivalent amount of compensated CO₂ by applying the established RF-based climate accounting relationship.

The operating scheme of the RF-meter is shown in Figure 1.

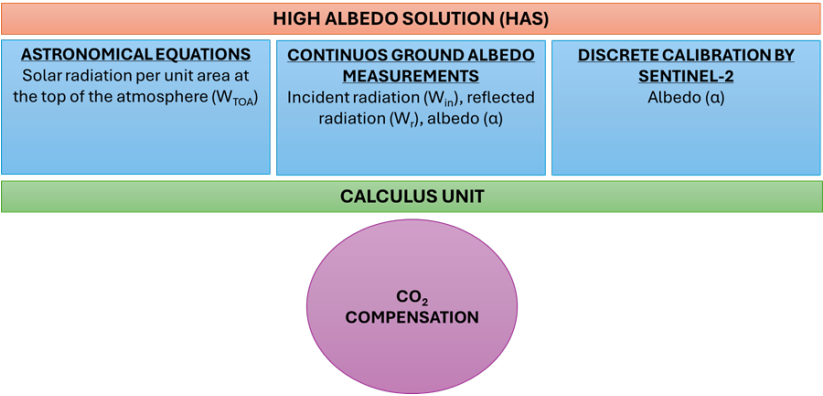


Figure 1. Radiative forcing (RF)-meter synoptic scheme

As illustrated in Figure 1, the evaluation of $RF(\Delta\alpha)$ requires two primary inputs: W_{TOA} , namely the incoming solar irradiance at the top of the atmosphere, derived from astronomical relationships; and W_{in} , namely the incident solar irradiance at ground level, measured continuously by means of albedometers.

Surface albedo is defined as the ratio between reflected solar radiation and incident solar radiation. Since ground level irradiance includes both direct and diffuse contributions, it has a hemispherical nature. By combining measured and calculated quantities, the RF associated with an albedo increment $\Delta\alpha$ can be expressed as follows (Eq. (1)):

$$RF_{\Delta\alpha} = -\frac{W_{in}^2}{W_{TOA}} \cdot \Delta\alpha = -W_{in} \cdot T_a \cdot \Delta\alpha \quad (1)$$

where, T_a denotes the atmospheric transmission coefficient for solar radiation. Eq. (1) derives from the surface radiative balance, under the assumption that the absorbed solar flux is proportional to ground incident irradiance W_{in} and to atmospheric transmissivity (T_a). When surface albedo increases, the absorbed fraction of incoming solar energy decreases accordingly, leading to the relationship expressed in Eq. (1). In this framework, a higher reflectance corresponds to a cooling contribution, i.e., to negative RF. This formulation is consistent with the physical interpretation adopted in previous studies on albedo induced RF [3, 5, 6].

To improve the robustness of the estimate, the $RF(\Delta\alpha)$ values obtained from Eq. (1) are calibrated using discrete Sentinel-2 satellite observations. For each overpass i , a calibration factor k_i is introduced and defined as the ratio between the satellite-derived albedo and the simultaneously measured ground albedo. The calibrated expression therefore becomes (Eq. (2)):

$$RF_{\Delta\alpha} = k_i \cdot W_{in} \cdot T_a \cdot \Delta\alpha \quad (2)$$

Finally, astronomical inputs, ground measurements, and satellite observations are processed within a calculation unit in order to estimate the CO_2 -equivalent compensation associated with HAS (Eq. (3)):

$$CO_{2,comp} = \frac{\sum_{i=1}^N \int_{T_i}^{T_{i+1}} S \cdot RF_{\Delta\alpha,i}(t) \cdot dt}{\int_0^T A_E \cdot k_T \cdot y_T(t) \cdot dt} \quad (3)$$

Eq. (3) expresses the cumulative climatic effect of the albedo change in terms of CO_2 -equivalent, according to an approach conceptually consistent with the IPCC Global Warming Potential framework.

A further methodological issue concerns the treatment of uncertainty. In real applications, the estimated forcing may be affected by sensor accuracy, temporal mismatch between ground measurements and satellite overpasses, atmospheric variability, and the progressive degradation of reflective properties caused by ageing, dust deposition, moisture, and surface wear. For this reason, the RF-meter should be interpreted not only as a calculation routine, but also as a measurement and verification framework in which calibration and long-term monitoring are essential. This perspective is particularly relevant if the method is intended to support future climate accounting or crediting applications, where traceability, repeatability, and durability of performance are critical requirements [8, 9].

The calculation unit was implemented in VBA. The algorithm applies Eqs. (2) and (3) over a discrete time step Δt and is organized into the following sections: Data, for importing albedometer measurements such as W_{in} , W_r , and α ; Settings, for defining constants and project parameters such as A_E , HAS area, and k_T , required for RF and CO_2 -offset estimation; Satellite, for entering the date, acquisition time, and albedo values associated with the j -th satellite overpass; and Calculation, for executing the computational routine and displaying the output results.

3 Governance Pathway for High-Albedo Solution Crediting

A possible pathway for the management of EC associated with HAS is outlined below. Its implementation would require both regulatory recognition and an appropriate organizational framework.

The proposed pathway may be structured in the following steps:

- 1) Formal recognition of the CO_2 -equivalent offset generated by HAS within climate-policy and regulatory discussions;
- 2) Possible inclusion of albedo based offset accounting within existing or future Emissions Trading Scheme;
- 3) Establishment of an independent technical body, here referred to as the Agency for Albedo, with the following responsibilities:
 - (a). Developing an operational protocol based on the RF-meter methodology;
 - (b). Evaluating HAS projects submitted by private individuals, companies, or public institutions;
 - (c). Validating projects that meet the required technical and monitoring criteria;
 - (d). Monitoring the time dependent CO_2 -equivalent offset associated with approved projects;

(e). Assigning EC annually on the basis of verified performance, for example according to a one credit per ton CO₂-equivalent criterion.

Within carbon-market frameworks, the economic value of one tonne of CO₂-equivalent is subject to market variability and regulatory evolution. Nevertheless, the existence of a recognized carbon price highlights the potential relevance of verified albedo-based mitigation actions. Under favourable climatic conditions, such as those found in high-irradiance regions, the cumulative CO₂-equivalent benefit per unit area may become significant, thereby giving reflective interventions a potentially measurable economic value within a crediting framework. Since many HAS applications can be implemented at relatively low cost, they may represent a cost-effective complement to other mitigation measures.

4 Potential Relevance for African High-Irradiance Regions

The RF-meter methodology is especially suitable for regions characterized by high solar irradiance and limited cloud cover, such as the North African belt. Preliminary calculations developed at CIRIAF (Interuniversity Research Centre on Pollution and Environment “Mauro Felli”, University of Perugia) suggest that, under such climatic conditions, relatively limited areas of highly reflective surface may generate significant CO₂-equivalent offsets [11, 12].

Beyond its technical application, the methodology may also be relevant in projects that combine climate mitigation with local development objectives. In this context, the “Albedo for Africa” initiative [13] has been proposed as an example of a strategy aimed at supporting sustainable development in Sub-Saharan regions through high-albedo interventions. The concept envisages the implementation of sustainable high-albedo villages in Sub-Saharan Africa, as schematically illustrated in Figure 2.

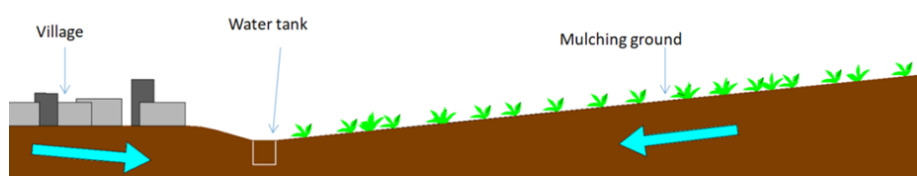


Figure 2. Scheme of sustainable high albedo village with high albedo parking, roofing, water recovery, and mulching ground

Within such a framework, the environmental and economic value associated with albedo enhancement could contribute to supporting local communities, while simultaneously promoting cost effective mitigation actions. These settlements may be organized around Homebuilt Housing Modules (HHMs) (Figure 3). Each HHM includes a residential unit equipped with essential services, photovoltaic panels for partial energy self-sufficiency, rainwater harvesting systems, wastewater recovery solutions, and reflective envelope components, such as high-albedo roofing.

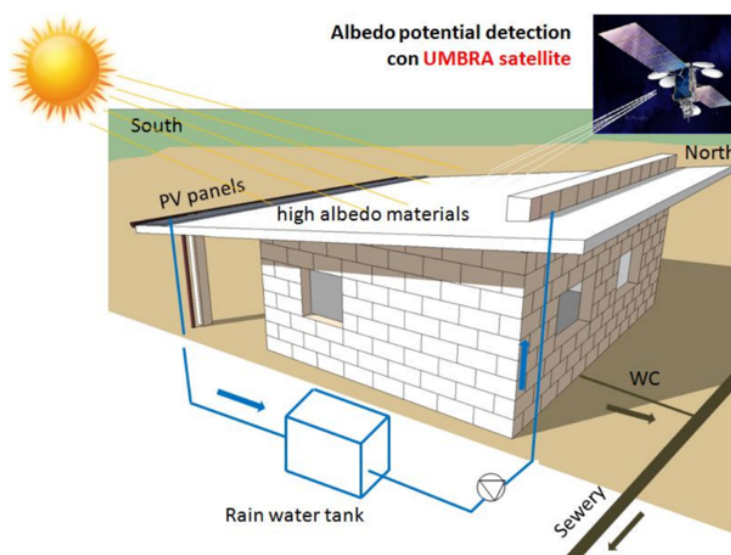


Figure 3. Scheme of Homebuilt Housing Module

From an economic standpoint, assigning value to the environmental benefits generated by these interventions could create development opportunities in low-income regions. Depending on the adopted financial assumptions, the long-term balance of such modules may become favorable without placing the full economic burden on local users.

In addition to the economic dimension, HHMs may provide broader social and environmental co-benefits, including improved self-sufficiency, greater awareness of environmental issues, and strengthened community resilience. More generally, this type of strategy may contribute to reducing summer cooling demand, mitigating local heat accumulation in densely populated areas, and supporting climate change mitigation through surface-based interventions. HAS can also be applied to infrastructure and agricultural systems. Typical examples include reflective parking areas (Figure 4) and high-reflectance mulching membranes in agriculture (Figure 5), which may combine radiative benefits with water saving effects and potential improvements in crop performance.



Figure 4. Specific application of High-Albedo Solution in infrastructures areas such parking area



Figure 5. High-reflective mulching membrane applied on the experimental field at CIRIAF Research Center, University of Perugia

Note: CIRIAF = Interuniversity Research Centre on Pollution and Environment “Mauro Felli”, University of Perugia.

The contribution of HAS to climate mitigation could become more relevant if reliable accounting and verification procedures allow these interventions to be considered within carbon crediting mechanisms. In this respect, the RF-meter is proposed as a technical tool that may facilitate the quantitative assessment of their climatic benefit.

5 Conclusions

The RF-meter is proposed as an operational framework for quantifying the CO₂-equivalent benefit associated with HAS. Its main purpose is to support a more robust technical assessment of reflective interventions in view of their possible consideration within EC schemes. The methodology combines continuous ground-based measurements, analytical calculations, and satellite observations. More specifically, it reconstructs the time dependent RF associated with an albedo increase produced by HAS. This makes it possible to express the climatic effect of the intervention in terms of compensated CO₂-equivalent, consistently with the RF-based accounting approach adopted in the literature.

The paper also outlines a possible governance pathway for the management of the resulting EC. Within this perspective, an independent technical body, referred to here as the Agency for Albedo, could be responsible for protocol definition, project evaluation, verification of time dependent offsets, and annual credit allocation on the basis of measured performance. Because many HAS applications can be implemented at relatively low cost, they may represent an economically accessible complement to other mitigation measures.

The methodology appears especially relevant in regions with high solar irradiance and limited cloud cover, such as the North African belt. In such contexts, the possible attribution of EC to verified albedo-based interventions could generate economic value while supporting local development opportunities. Overall, the RF-meter may provide a useful technical basis for linking reflective surface interventions with climate mitigation and, potentially, with future carbon accounting frameworks.

Author Contributions

Conceptualization, F.R.; methodology, F.R.; software, A.D.G.; validation, F.R. and A.D.G.; formal analysis, A.D.G.; investigation, F.R. and A.D.G.; resources, F.R.; data curation, A.D.G.; writing—original draft preparation, A.D.G.; writing—review and editing, F.R.; visualization, A.D.G.; supervision, F.R.; project administration, F.R.; funding acquisition, F.R. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

A_E	Earth surface
EC	Emission credit
HAS	High-Albedo Solution
HHM	Homebuilt Housing Module
IPCC	International Panel on Climate Change
k_i	Ratio between the satellite albedo at the i -th overpass and the simultaneous ground albedo
k_r	Radiative efficiency of CO ₂
N	Number of satellite passages during T
RF	Radiative forcing
$RF(\Delta\alpha)$	Radiative forcing due to an albedo increase
S	Area of the HAS
T_a	Solar energy transmission coefficient in the atmosphere
T_i and T_{i+1}	Times of the i -th and $(i + 1)$ -th satellite passes
W_{in}	Incident radiation
W_{TOA}	Incoming solar radiation at the top of the atmosphere
$y_r(t)$	Time-dependent decay in abundance of CO ₂ following an instantaneous release of it at time $t = 0$